

# Multiple Choice Questions

Machine Learning | Python | NLP | Azure AI | Data Science

Total Questions: 172

**Q1. In machine learning, what term refers to a situation where a model learns the training data too well, capturing noise and leading to poor generalization on new data?**

- (a) Bias-Variance Tradeoff
- (b) Regularization
- (c) Underfitting
- (d) Overfitting ✓

**Q2. What is the purpose of name mangling in Python classes?**

- (a) Obfuscating code
- (b) Creating unique variable names ✓
- (c) Improving code performance
- (d) Preventing access to class attributes

**Q3. What is the primary purpose of using nested loops in programming?**

- (a) To reduce code complexity
- (b) To create recursive functions
- (c) To execute a loop inside another loop ✓
- (d) To improve code readability

**Q4. In Python, what is the purpose of file handling?**

- (a) To delete unnecessary files
- (b) To control the display of files in the terminal
- (c) To create backups of important files
- (d) To read from and write to files on disk ✓

**Q5. When integrating Azure AI services into a CI/CD pipeline, which practice involves automating the deployment of AI models and services to production environments?**

- (a) CI/CD Integration ✓
- (b) Data Collection
- (c) Model Training
- (d) Feature Engineering

**Q6. In NLP, what is the process of reducing words to their base or dictionary form while ensuring they are still valid words?**

- (a) Lemmatization ✓
- (b) Tokenization
- (c) Chunking
- (d) Stemming

**Q7. In statistics, which term refers to the function that describes the likelihood of various outcomes in a random experiment?**

- (a) Probability distribution ✓
- (b) Hypothesis testing
- (c) Central limit theorem
- (d) Normal distribution

**Q8. Which measure provides an indication of how spread out the values in a dataset are?**

- (a) Z-Score
- (b) Coefficient of Variation
- (c) Variance
- (d) Standard Deviation ✓

**Q9. What is the primary purpose of string operations in programming?**

- (a) To generate random numbers
- (b) To control robotic movements
- (c) To create complex mathematical equations
- (d) To manipulate and process text ✓

**Q10. What is the process of breaking a text into individual words or tokens?**

- (a) Lemmatization
- (b) Sentence tokenization
- (c) Word tokenization ✓
- (d) Text normalization

**Q11. Which command is used to display the contents of a directory in the terminal?**

- (a) ls ✓
- (b) display
- (c) dir
- (d) show

**Q12. In univariate linear regression with vectorization, what advantage does vectorization offer over using loops for computations?**

- |     |                                     |
|-----|-------------------------------------|
| (a) | Reduced memory usage                |
| (b) | Easier code debugging               |
| (c) | Improved computational efficiency ✓ |
| (d) | Enhanced model interpretability     |

**Q13. Which deep learning framework is developed by Google and provides a comprehensive suite of tools and libraries for building and training neural networks?**

- |     |              |
|-----|--------------|
| (a) | Keras        |
| (b) | PyTorch      |
| (c) | TensorFlow ✓ |
| (d) | Theano       |

**Q14. Which service can you use to analyze images and extract text from them in Microsoft Azure?**

- |     |                     |
|-----|---------------------|
| (a) | Anomaly Detector    |
| (b) | Personalizer        |
| (c) | Metrics Advisor     |
| (d) | Content Moderator ✓ |

**Q15. In Pandas, which type of join combines rows from two DataFrames based on common columns, keeping only the matching rows?**

- |     |              |
|-----|--------------|
| (a) | Left join    |
| (b) | Right join   |
| (c) | Inner join ✓ |
| (d) | Outer join   |

**Q16. What type of solution can you create using Microsoft Azure's Conversational AI services?**

- |     |                         |
|-----|-------------------------|
| (a) | Speech recognition ✓    |
| (b) | Image classification    |
| (c) | Decision support system |
| (d) | Text analysis           |

**Q17. Which Azure service provides features like text-to-speech, translation, and focus mode, enhancing readability for users with varying abilities?**

- (a) Azure Personalizer
- (b) Azure Content Moderator
- (c) Azure Text Analytics
- (d) Azure Immersive Reader ✓

**Q18. In object-oriented programming, what is an abstract method?**

- (a) A method that can't be called
- (b) A method that is automatically generated
- (c) A method that has no implementation ✓
- (d) A method that can't be overridden

**Q19. Which measures assess the spread or dispersion of data points around the mean?**

- (a) Mean and Standard Deviation
- (b) Mean and Median
- (c) Variance and Standard Deviation ✓
- (d) Range and Mode

**Q20. Which measure indicates how many standard deviations a data point is away from the mean?**

- (a) Quartile
- (b) Z-Score ✓
- (c) Range
- (d) Percentile

**Q21. What does broadcasting in Numpy allow you to do?**

- (a) Convert arrays to different data types
- (b) Perform element-wise operations on arrays with different shapes ✓
- (c) Combine arrays into a single array
- (d) Resize arrays to match the largest array

**Q22. In Linux, which command is used to update the package list?**

- (a) apt-get ✓
- (b) apt-upgrade
- (c) apt-list
- (d) apt-update

**Q23. In machine learning, which algorithm is used for both classification and regression tasks and aims to find the hyperplane that best separates data points of different classes while maximizing the margin?**

- (a) Linear Regression
- (b) Support Vector Machine (SVM) ✓
- (c) Decision Tree
- (d) K-Means Clustering

**Q24. What is the main goal of machine learning?**

- (a) To program computers to think like humans
- (b) To enable computers to learn from data and improve performance over time ✓
- (c) To automate tasks without using data
- (d) To optimize computer hardware

**Q25. Which Numpy function is used to create a copy of an array?**

- (a) `numpy.clone()`
- (b) `numpy.copy()` ✓
- (c) `numpy.duplicate()`
- (d) `numpy.replicate()`

**Q26. What is an inner/nested class in OOP?**

- (a) A class defined within another class ✓
- (b) A class that can only be instantiated once
- (c) A class that inherits from another class
- (d) A class with no methods or attributes

**Q27. What is the measure of central tendency that represents the average value of a dataset?**

- (a) Mode
- (b) Median
- (c) Mean ✓
- (d) Range

**Q28. What method is used to select elements from a Numpy array based on a condition?**

- (a) `conditional_selection()`
- (b) `filter()`
- (c) `logical_select()`
- (d) `where()` ✓

**Q29. In natural language processing, what is the process of grouping adjacent words together into meaningful phrases or 'chunks' based on their part-of-speech tags?**

- (a) Chunking ✓
- (b) NER tagging
- (c) Chinking
- (d) POS tagging

**Q30. In statistics, which term describes a variable that can take on distinct, separate values?**

- (a) Quantitative
- (b) Qualitative
- (c) Discrete ✓
- (d) Continuous

**Q31. Which library is commonly used for natural language processing tasks such as tokenization, stemming, and part-of-speech tagging?**

- (a) Matplotlib
- (b) SpaCy
- (c) Scikit-learn
- (d) NLTK ✓

**Q32. Which Seaborn function is used to create a bar plot of categorical data?**

- (a) sns.catplot()
- (b) sns.barplot() ✓
- (c) sns.scatterplot()
- (d) sns.lineplot()

**Q33. What is a common step in data pre-processing for NLP that involves converting all characters to lowercase?**

- (a) Lemmatization
- (b) Stopword removal
- (c) Tokenization
- (d) Lowercasing ✓

**Q34. In natural language processing, what mechanism is used to selectively focus on different parts of input sequences while making predictions?**

- (a) Bag-of-Words
- (b) Attention Mechanism ✓
- (c) Named Entity Recognition
- (d) Sequence-to-Sequence Model

**Q35. In neural networks, what type of functions are applied to the output of neurons to introduce nonlinearity and enable the network to learn complex relationships in data?**

- (a) Gradient functions
- (b) Activation functions ✓
- (c) Loss functions
- (d) Regularization functions

**Q36. What field of study focuses on enabling computers to understand, interpret, and generate human language?**

- (a) Computer Vision
- (b) Data Science
- (c) Natural Language Processing ✓
- (d) Machine Learning

**Q37. Which Azure service provides language translation capabilities and supports a wide range of languages for translation?**

- (a) Azure Text Analytics
- (b) Azure Content Moderator
- (c) Azure Speech Service
- (d) Azure Translator ✓

**Q38. Which data structure is commonly used in Pandas to store tabular data?**

- (a) Arrays
- (b) Lists
- (c) DataFrames ✓
- (d) Sets

**Q39. Which measure quantifies the relative variability of a dataset, considering the standard deviation and mean?**

- (a)** Coefficient of Variation ✓
- (b)** Skewness
- (c)** Kurtosis
- (d)** Variance

**Q40. In neural networks, what graphical representation visualizes the sequence of mathematical operations performed during forward and backward passes?**

- (a)** Loss Function
- (b)** Gradient Descent
- (c)** Activation Function
- (d)** Computational Graph ✓

**Q41. When using a cloud service, what is the URL or network address that serves as the entry point for accessing the service's features and functionalities?**

- (a)** API Key
- (b)** Default Endpoint ✓
- (c)** Service Manifest
- (d)** Authentication Token

**Q42. Which command is used to change the permissions of a file in Unix-like systems?**

- (a)** touch
- (b)** kill
- (c)** chmod ✓
- (d)** ps

**Q43. In natural language processing, which model aims to predict a target word based on its surrounding context words?**

- (a)** Continuous Bag-of-Words (CBOW) ✓
- (b)** Named Entity Recognition
- (c)** Part-of-Speech Tagging
- (d)** Skip-gram

**Q44. What is dynamic polymorphism in object-oriented programming?**

- |     |                                                                   |
|-----|-------------------------------------------------------------------|
| (a) | A technique to combine multiple classes into one                  |
| (b) | The ability of a subclass to override methods of its base class ✓ |
| (c) | A way to prevent inheritance between classes                      |
| (d) | The process of defining new data types                            |

**Q45. In the context of data, what does 'structured' data refer to?**

- |     |                                                           |
|-----|-----------------------------------------------------------|
| (a) | Data that is randomly generated                           |
| (b) | Data that is not organized in any specific way            |
| (c) | Data that is not useful for analysis                      |
| (d) | Data that is organized into a specific format or schema ✓ |

**Q46. When developing a computer vision solution that requires real-time object detection and recognition, which cloud service is most suitable?**

- |     |                               |
|-----|-------------------------------|
| (a) | Amazon Rekognition            |
| (b) | Google Cloud Vision           |
| (c) | Azure Cognitive Services ✓    |
| (d) | IBM Watson Visual Recognition |

**Q47. In univariate linear regression without vectorization, how is the gradient of the cost function calculated for the slope parameter?**

- |     |                                                |
|-----|------------------------------------------------|
| (a) | Using matrix multiplication                    |
| (b) | Using a direct formula                         |
| (c) | Using a loop to iterate over each data point ✓ |
| (d) | By taking the derivative of the cost function  |

**Q48. Which Seaborn function is used to create a heatmap to visualize the correlation matrix of a DataFrame?**

- |     |                       |
|-----|-----------------------|
| (a) | sns.corrplot()        |
| (b) | sns.matrixplot()      |
| (c) | sns.correlationplot() |
| (d) | sns.heatmap() ✓       |

**Q49. Which symbol is used in a regular expression to match any single character?**

- |     |     |
|-----|-----|
| (a) | ^   |
| (b) | *   |
| (c) | . ✓ |
| (d) | ?   |

**Q50. What are universal array functions in Numpy?**

- |     |                                                 |
|-----|-------------------------------------------------|
| (a) | Functions that perform matrix multiplication    |
| (b) | Functions that operate element-wise on arrays ✓ |
| (c) | Functions that only work on 1D arrays           |
| (d) | Functions that require external libraries       |

**Q51. When deploying a machine learning model as a web service in Microsoft Azure, which service allows you to create, deploy, and manage RESTful APIs for your trained models?**

- |     |                                  |
|-----|----------------------------------|
| (a) | Azure Cognitive Services         |
| (b) | Azure Functions                  |
| (c) | Azure Logic Apps                 |
| (d) | Azure Machine Learning Service ✓ |

**Q52. What field of study focuses on enabling computers to understand, interpret, and generate human language? (II)**

- |     |                               |
|-----|-------------------------------|
| (a) | Computer Vision               |
| (b) | Data Science                  |
| (c) | Natural Language Processing ✓ |
| (d) | Machine Learning              |

**Q53. When planning a container deployment for your application, which technology allows you to package and distribute applications along with their dependencies?**

- |     |                     |
|-----|---------------------|
| (a) | Docker Containers ✓ |
| (b) | Azure Functions     |
| (c) | Kubernetes          |
| (d) | Virtual Machines    |

**Q54. What technique is used in Pandas to handle missing data in a DataFrame?**

- |     |                   |
|-----|-------------------|
| (a) | Data encryption   |
| (b) | Data imputation ✓ |
| (c) | Data dropping     |
| (d) | Data shuffling    |

**Q55. Which technique in natural language processing is used to convert words into dense vector representations, capturing semantic meanings and relationships between words?**

- (a) Word2Vec ✓
- (b) Named Entity Recognition
- (c) Bag-of-Words
- (d) POS tagging

**Q56. What is the main goal of using exception handling in programming?**

- (a) To make code shorter
- (b) To gracefully handle errors and exceptions ✓
- (c) To make code run faster
- (d) To prevent any errors from occurring

**Q57. In natural language processing, what is the process of identifying and classifying named entities in text?**

- (a) NER tagging ✓
- (b) POS tagging
- (c) Stemming
- (d) Tokenization

**Q58. In Pandas, which operation is used to perform element-wise addition of two DataFrames with matching indexes and columns?**

- (a) df.concat()
- (b) df.add() ✓
- (c) df.combine()
- (d) df.merge()

**Q59. In object-oriented programming, what is multilevel inheritance?**

- (a) Inheriting from multiple parent classes
- (b) Inheriting from a child class
- (c) Inheriting from a grandparent class ✓
- (d) Inheriting from a single parent class

**Q60. In Pandas, which method is used to save a DataFrame to a CSV file?**

- (a) df.write\_csv()
- (b) df.export\_csv()
- (c) df.to\_csv() ✓
- (d) df.save\_csv()

**Q61. What is the primary application of natural language processing (NLP) solutions?**

- |     |                                   |
|-----|-----------------------------------|
| (a) | Image analysis                    |
| (b) | Video compression                 |
| (c) | Text analysis and understanding ✓ |
| (d) | Audio processing                  |

**Q62. In machine learning, what is the process of selecting a subset of relevant features to improve model performance and reduce complexity?**

- |     |                     |
|-----|---------------------|
| (a) | Feature extraction  |
| (b) | Feature selection ✓ |
| (c) | NER tagging         |
| (d) | Tokenization        |

**Q63. Which measure of central tendency is most affected by outliers?**

- |     |                    |
|-----|--------------------|
| (a) | Standard Deviation |
| (b) | Mean ✓             |
| (c) | Median             |
| (d) | Mode               |

**Q64. Which Azure service provides the capability to moderate and classify content to ensure it meets appropriate standards?**

- |     |                           |
|-----|---------------------------|
| (a) | Azure Cognitive Services  |
| (b) | Azure Content Moderator ✓ |
| (c) | Azure Machine Learning    |
| (d) | Azure Computer Vision     |

**Q65. In natural language processing, which model aims to predict a target word based on its surrounding context words? (II)**

- |     |                                  |
|-----|----------------------------------|
| (a) | Continuous Bag-of-Words (CBOW) ✓ |
| (b) | Named Entity Recognition         |
| (c) | Part-of-Speech Tagging           |
| (d) | Skip-gram                        |

**Q66. What is the advantage of using functions in programming?**

- |     |                                                              |
|-----|--------------------------------------------------------------|
| (a) | They allow you to define new data types                      |
| (b) | They help you optimize memory usage                          |
| (c) | They allow you to create GUI interfaces                      |
| (d) | They allow you to reuse code and improve code organization ✓ |

**Q67. What is a lexical database in NLP that provides a semantic network of words and their relationships, including synonyms and hypernyms?**

- (a) NER tagging
- (b) Stemming
- (c) POS tagging
- (d) WordNet ✓

**Q68. What is the primary purpose of Numpy in Python?**

- (a) To create graphical user interfaces
- (b) To perform mathematical operations on arrays and matrices ✓
- (c) To manipulate strings
- (d) To generate random numbers

**Q69. In deep neural networks, what problem arises when gradients during backpropagation become very small, causing slow convergence?**

- (a) Overfitting
- (b) Vanishing gradients ✓
- (c) Exploding gradients
- (d) Underfitting

**Q70. In sequence models, which type of RNN architecture processes input sequences both in forward and backward directions?**

- (a) Bi-Directional LSTM/RNN ✓
- (b) CNN
- (c) Uni-Directional LSTM/RNN
- (d) Transformer

**Q71. How does the Anomaly Detector service in Microsoft Azure work?**

- (a) It identifies abnormalities based on statistical analysis
- (b) It compares data to pre-defined patterns
- (c) It relies on user-defined rules to detect anomalies
- (d) It uses machine learning to predict anomalies ✓

**Q72. Which Azure service provides text analytics capabilities for language detection, sentiment analysis, and named entity recognition?**

- (a) Azure Content Moderator
- (b) Azure Speech Service
- (c) Azure Text Analytics ✓
- (d) Azure Translator

**Q73. In data analysis, what process involves converting continuous numerical data into discrete intervals or bins?**

- (a) Binning ✓
- (b) Discretization
- (c) Categorization
- (d) Segmentation

**Q74. Which type of data includes non-numerical information such as labels or categories?**

- (a) Numerical data
- (b) Qualitative data
- (c) Categorical data ✓
- (d) Quantitative data

**Q75. Which command is used to display the contents of a file in a Unix-like operating system?**

- (a) read
- (b) view
- (c) cat ✓
- (d) open

**Q76. In programming, what is the term for a function defined inside another function?**

- (a) Nested function ✓
- (b) Child function
- (c) Inner function
- (d) Subfunction

**Q77. Which Numpy function is used to perform element-wise addition of two arrays?**

- (a) numpy.add() ✓
- (b) numpy.append()
- (c) numpy.sum()
- (d) numpy.plus()

**Q78. In machine learning, which loss function is commonly used for classification tasks and measures the dissimilarity between predicted probabilities and actual class labels?**

- (a) Mean Absolute Error
- (b) Mean Squared Error
- (c) Huber Loss
- (d) Cross-Entropy ✓

**Q79. Which type of sequence model is used for tasks like machine translation?**

- (a) Feedforward Neural Networks
- (b) Recurrent Neural Networks (RNN) ✓
- (c) Convolutional Neural Networks (CNN)
- (d) Decision Trees

**Q80. Which command is used to change the permissions of a file in Unix-like systems? (II)**

- (a) touch
- (b) kill
- (c) chmod ✓
- (d) ps

**Q81. What is the primary purpose of Artificial Intelligence (AI)?**

- (a) To simulate human emotions.
- (b) To play board games.
- (c) To create physical robots only.
- (d) To perform tasks without any human intervention. ✓

**Q82. When managing authentication for a cloud resource, which approach involves configuring IAM policies to control who can access and perform actions on the resource?**

- (a) IAM Policies ✓
- (b) OAuth Tokens
- (c) API Keys
- (d) Encryption Keys

**Q83. Which keyword is used in Python to introduce an alternative condition if the initial condition is false?**

- (a) Elself
- (b) Otherwise
- (c) Else ✓
- (d) Unless

**Q84. In univariate linear regression with gradient descent, what is the role of the gradient descent algorithm?**

- |     |                                                             |
|-----|-------------------------------------------------------------|
| (a) | To calculate the slope of the regression line               |
| (b) | To maximize the cost function and find optimal parameters   |
| (c) | To minimize the cost function and find optimal parameters ✓ |
| (d) | To calculate the intercept of the regression line           |

**Q85. In neural networks, what are the two main steps involved in the training process (computing output and updating weights)?**

- |     |                                            |
|-----|--------------------------------------------|
| (a) | Data Preprocessing and Feature Selection   |
| (b) | Forward and Backward Passes ✓              |
| (c) | Hyperparameter Tuning and Cross-Validation |
| (d) | Model Compilation and Evaluation           |

**Q86. In Python, which library provides an easy-to-use interface for building and training neural networks?**

- |     |              |
|-----|--------------|
| (a) | Keras ✓      |
| (b) | TensorFlow   |
| (c) | Scikit-learn |
| (d) | PyTorch      |

**Q87. Which measure quantifies the strength and direction of a linear relationship between two variables?**

- |     |                           |
|-----|---------------------------|
| (a) | Mean                      |
| (b) | Correlation Coefficient ✓ |
| (c) | Standard Deviation        |
| (d) | Variance                  |

**Q88. What is the purpose of list comprehension in Python?**

- |     |                                                         |
|-----|---------------------------------------------------------|
| (a) | To create complex nested lists                          |
| (b) | To manipulate and filter lists using a concise syntax ✓ |
| (c) | To write long and verbose loops                         |
| (d) | To generate random numbers                              |

**Q89. Which linguistic component focuses on the structure and rules governing the arrangement of words and phrases to form sentences?**

- (a) Discourse
- (b) Semantics
- (c) Syntax ✓
- (d) Pragmatics

**Q90. In programming, what is the purpose of a function?**

- (a) To organize code into reusable blocks ✓
- (b) To create loops
- (c) To print output
- (d) To store data

**Q91. What type of data consists of numerical values that represent quantities?**

- (a) Binary data
- (b) Categorical data
- (c) Qualitative data
- (d) Quantitative data ✓

**Q92. Which measure provides information about the relative position of a data point within a dataset?**

- (a) Mean
- (b) Standard Deviation
- (c) Percentile ✓
- (d) Range

**Q93. In probability theory, what type of probability considers the likelihood of events occurring together?**

- (a) Joint probability ✓
- (b) Conditional probability
- (c) Independent probability
- (d) Marginal probability

**Q94. In supervised machine learning, what is the primary task of the model?**

- (a) To generate new data samples
- (b) To preprocess the data
- (c) To cluster similar data points
- (d) To learn from labeled training data to make predictions on new, unseen data ✓

**Q95. Which type of plot displays data for a single variable, showing its distribution and characteristics?**

- (a) Bivariate plot
- (b) Multivariate plot
- (c) Scatter plot
- (d) Univariate plot ✓

**Q96. Which cloud service would be a suitable choice for NLP tasks including sentiment analysis, named entity recognition, and language translation?**

- (a) Google Cloud Natural Language
- (b) Azure Cognitive Services ✓
- (c) IBM Watson Natural Language Understanding
- (d) Amazon Comprehend

**Q97. What is hierarchical inheritance in OOP?**

- (a) Inheriting from a single parent class ✓
- (b) Inheriting from a grandparent class
- (c) Inheriting from multiple child classes
- (d) Inheriting from multiple parent classes

**Q98. When planning for an AI solution aligned with Responsible AI, which practice ensures transparency and explainability of AI models?**

- (a) Model Accuracy
- (b) Explainability and Transparency ✓
- (c) Performance Metrics
- (d) Data Collection

**Q99. In logistic regression for multiclass classification, what type of decision boundary is typically learned by the model?**

- (a) Exponential decision boundary
- (b) Linear decision boundary ✓
- (c) Non-linear decision boundary
- (d) Constant decision boundary

**Q100. In machine learning, which loss function is commonly used for classification tasks and measures dissimilarity between predicted probabilities and actual class labels? (II)**

- (a) Mean Absolute Error
- (b) Mean Squared Error
- (c) Huber Loss
- (d) Cross-Entropy ✓

**Q101. In Python, which function is used to apply a function to all elements in an iterable?**

- (a) apply()
- (b) filter()
- (c) iterate()
- (d) map() ✓

**Q102. What is a lexical database in NLP that provides a semantic network of words and relationships? (II)**

- (a) NER tagging
- (b) Stemming
- (c) POS tagging
- (d) WordNet ✓

**Q103. In deep learning, which type of neural network is commonly used for image classification tasks, leveraging 2D convolutional layers?**

- (a) 2D Convolutional Neural Network (CNN) ✓
- (b) Autoencoder
- (c) Multi-Layer Perceptron (MLP)
- (d) Recurrent Neural Network (RNN)

**Q104. In Pandas, what operation allows you to group a DataFrame based on a column and then perform aggregate functions on each group?**

- (a) GroupBy ✓
- (b) Aggregate
- (c) Group
- (d) Combine

**Q105. When configuring security for Microsoft Azure AI services, which service enables you to manage identity and access for applications?**

- (a) Azure Key Vault
- (b) Azure Security Center
- (c) Azure Firewall
- (d) Azure Active Directory ✓

**Q106. What is the purpose of 'magic functions' or 'dunder methods' in Python?**

- (a) Enforcing strict naming conventions
- (b) Defining private methods in a class
- (c) Creating unique variable names
- (d) Overloading operators and providing special behavior ✓

**Q107. In Microsoft Azure, which service provides a platform for building, training, and deploying machine learning models?**

- (a) Azure Machine Learning ✓
- (b) Azure AI Builder
- (c) Azure Data Factory
- (d) Azure Cognitive Services

**Q108. In machine learning, what type of problem involves predicting a continuous numerical output?**

- (a) Regression ✓
- (b) Classification
- (c) Reinforcement learning
- (d) Clustering

**Q109. Which Azure Cognitive Services offering is best suited for converting spoken language into text, text-to-speech, and speaker recognition?**

- (a) Azure Text Analytics
- (b) Azure Language Understanding
- (c) Azure Speech Service ✓
- (d) Azure Translator

**Q110. When creating a new resource in Microsoft Azure, which web-based interface provides a GUI to select and configure services?**

- (a) Azure PowerShell
- (b) Azure Portal ✓
- (c) Azure CLI
- (d) Azure SDK

**Q111. In programming, what is the term for a function defined inside another function? (II)**

- (a) Nested function ✓
- (b) Child function
- (c) Inner function
- (d) Subfunction

**Q112. What is an object in object-oriented programming?**

- (a) A block of code that is executed once
- (b) A function that defines a class
- (c) An instance of a class with its own properties and methods ✓
- (d) A value that can be changed during runtime

**Q113. In natural language processing, what is the process of reducing a word to its base or root form by removing suffixes or prefixes?**

- (a) Segmentation
- (b) Stemming ✓
- (c) Lemmatization
- (d) Tokenization

**Q114. Which Seaborn function is used to create a histogram with a kernel density estimate (KDE) plot?**

- (a) sns.distplot() ✓
- (b) sns.histplot()
- (c) sns.kdeplot()
- (d) sns.plot.hist()

**Q115. In object-oriented programming, what are class variables?**

- (a) Variables that are shared among all instances of a class ✓
- (b) Variables that belong to the class itself
- (c) Variables that store data unique to each instance of a class
- (d) Variables used for method definition

**Q116. What is the purpose of selecting the appropriate Microsoft Azure AI service for a solution?**

- (a) To simplify the code development process ✓
- (b) To ensure the responsible use of AI technologies
- (c) To maximize profits from AI solutions
- (d) To showcase various AI services to clients

**Q117. In languages without explicit spaces between words, what is the process of dividing a sequence of characters into individual words called?**

- (a) Sentence tokenization
- (b) Part-of-speech tagging
- (c) Text normalization
- (d) Word segmentation ✓

**Q118. In Pandas, which data structure is used to store one-dimensional labeled data, similar to a column in a spreadsheet?**

- (a) List
- (b) DataFrame
- (c) Array
- (d) Series ✓

**Q119. Seaborn is a Python data visualization library based on which other library?**

- (a) Scikit-learn
- (b) Matplotlib ✓
- (c) Pandas
- (d) Numpy

**Q120. Which type of plot displays data for a single variable, showing its distribution and characteristics? (II)**

- (a) Bivariate plot
- (b) Multivariate plot
- (c) Scatter plot
- (d) Univariate plot ✓

**Q121. In data analysis, what process involves converting continuous numerical data into discrete intervals or bins? (II)**

- (a) Binning ✓
- (b) Discretization
- (c) Categorization
- (d) Segmentation

**Q122. In machine learning, which ensemble method combines multiple decision trees by training them on different subsets of the training data?**

- (a) Random Forest ✓
- (b) K-Means Clustering
- (c) Gradient Boosting
- (d) AdaBoost

**Q123. How can you save a Numpy array to a file?**

- (a) `numpy.save()` ✓
- (b) `numpy.store()`
- (c) `numpy.write()`
- (d) `numpy.export()`

**Q124. In deep learning, what type of neural network architecture consists of multiple layers of interconnected neurons where information flows only in one direction?**

- (a) Recurrent Neural Network (RNN)
- (b) Long Short-Term Memory (LSTM)
- (c) MLP Feed Forward Neural Network ✓
- (d) Convolutional Neural Network (CNN)

**Q125. What does Bayesian probability involve?**

- (a) Estimating probabilities using random sampling
- (b) Counting the number of occurrences in a dataset
- (c) Calculating probabilities based on prior knowledge and observations ✓
- (d) Using logical operators to find probabilities

**Q126. How can you create a Numpy array from a Python list?**

- (a) `numpy.create()`
- (b) `numpy.array()` ✓
- (c) `numpy.from_list()`
- (d) `numpy.list()`

**Q127. Which ensemble method combines weak learners sequentially, where each subsequent learner focuses more on misclassified data?**

- (a) K-Means Clustering
- (b) Random Forest
- (c) Bagging
- (d) Boosting ✓

**Q128. In Microsoft Azure, what networking solution allows you to isolate and secure your cloud resources by creating private network connections?**

- (a) Azure Active Directory
- (b) Azure Firewall
- (c) Azure Virtual Networks ✓
- (d) Azure Security Center

**Q129. In machine learning, what term refers to a situation where one class has significantly more instances than other classes in a dataset?**

- (a) Overfitting
- (b) Dataset augmentation
- (c) Underfitting
- (d) Dataset imbalance ✓

**Q130. Which Azure service provides language translation capabilities and supports a wide range of languages? (II)**

- (a) Azure Text Analytics
- (b) Azure Content Moderator
- (c) Azure Speech Service
- (d) Azure Translator ✓

**Q131. Which type of loop is suitable when the number of iterations is known beforehand?**

- (a) While loop
- (b) Do-while loop
- (c) For loop ✓
- (d) Infinite loop

**Q132. What is inheritance in object-oriented programming?**

- (a) It allows a class to inherit properties and methods from another class ✓
- (b) It's a way to create new classes from scratch
- (c) It's a way to define unrelated classes
- (d) It's a process of copying and pasting code from one class to another

**Q133. In the Bag-of-Words (BoW) model, how is a text represented as a feature vector for NLP tasks?**

- (a) By using lemmatization
- (b) By counting the frequency of words in the text ✓
- (c) By removing stopwords
- (d) By converting words to lowercase

**Q134. What are the fundamental building blocks of programming that manipulate data?**

- (a) Operators ✓
- (b) Keywords
- (c) Variables
- (d) Functions

**Q135. When creating an Azure AI resource, which service in the Azure portal allows you to provision and manage AI capabilities?**

- (a) Azure App Service
- (b) Azure Storage
- (c) Azure Virtual Machine
- (d) Azure Cognitive Services ✓

**Q136. Which Azure service offers automatic speech recognition capabilities for converting spoken language into text data?**

- (a) Azure Translator
- (b) Azure Text Analytics
- (c) Azure Speech Service ✓
- (d) Azure Content Moderator

**Q137. When developing an application that requires custom image classification and object detection, which Azure Cognitive Service provides a user-friendly platform to train custom models?**

- (a) Azure Computer Vision
- (b) Custom Vision ✓
- (c) Azure Content Moderator
- (d) Azure Video Indexer

**Q138. What does polymorphism in OOP refer to?**

- (a) Overriding parent class methods
- (b) Allowing operators to work with different data types objects as instances of a common superclass
- (c) Creating complex class hierarchies
- (d) Treating different class objects as instances of a common superclass ✓

**Q139. Which Azure Cognitive Services offering is best suited for speech-to-text, text-to-speech, and speaker recognition? (II)**

- (a) Azure Text Analytics
- (b) Azure Language Understanding
- (c) Azure Speech Service ✓
- (d) Azure Translator

**Q140. In NLP, what is the process of converting text to a consistent format, such as converting all characters to lowercase and removing special characters?**

- (a) Stemming
- (b) Part-of-speech tagging
- (c) Tokenization
- (d) Text normalization ✓

**Q141. Which service in the Azure portal allows you to provision and manage AI capabilities such as cognitive services and machine learning models? (II)**

- (a) Azure App Service
- (b) Azure Storage
- (c) Azure Virtual Machine
- (d) Azure Cognitive Services ✓

**Q142. Which data structure is used to store an unordered collection of unique elements in Python?**

- (a) Arrays
- (b) Sets ✓
- (c) Lists
- (d) Tuples

**Q143. In natural language processing, what is the process of assigning grammatical tags to words in a text based on their roles in a sentence?**

- (a) POS tagging ✓
- (b) Text normalization
- (c) Stemming
- (d) Tokenization

**Q144. In natural language processing, what is the process of reducing a word to its base or root form by removing suffixes or prefixes? (II)**

- (a) Segmentation
- (b) Stemming ✓
- (c) Lemmatization
- (d) Tokenization

**Q145. In supervised machine learning, which component represents the function that the model is trying to learn and approximate?**

- (a) Optimizer
- (b) Hypothesis ✓
- (c) Cost function
- (d) Labeled data

**Q146. What are the fundamental building blocks of programming that manipulate data? (II)**

- (a) Operators ✓
- (b) Keywords
- (c) Variables
- (d) Functions

**Q147. What is the purpose of using keyword arguments in function calls?**

- (a) To increase the performance of a function
- (b) To assign a name to an argument ✓
- (c) To change the order of arguments
- (d) To make a function private

**Q148. What is the purpose of software installation tools like Anaconda and VSCode in AI development?**

- (a) To demonstrate origami techniques
- (b) To install and manage necessary programming environments ✓
- (c) To showcase gardening skills
- (d) To create virtual reality experiences

**Q149. What is the primary application of Convolutional Neural Networks (CNNs)?**

- (a) Tabular data analysis
- (b) Text classification
- (c) Image and video analysis ✓
- (d) Speech recognition

**Q150. Which type of words are commonly removed from text during pre-processing due to their high frequency and lack of significant meaning?**

- (a) Verbs
- (b) Stopwords ✓
- (c) Adjectives
- (d) Punctuation

**Q151. Which cloud service offers a comprehensive Text Analytics API for extracting insights from unstructured text data?**

- (a) Azure Cognitive Services ✓
- (b) IBM Watson
- (c) Google Cloud AI
- (d) Amazon SageMaker

**Q152. What does the term 'probability' refer to in statistics?**

- (a) The ratio of successful outcomes to the total possible outcomes ✓
- (b) The number of trials in an experiment
- (c) The certainty of an event occurring
- (d) The number of times an event occurs

**Q153. Which Azure Cognitive Service is designed to understand user preferences and deliver customized experiences based on real-time interaction data?**

- (a) Azure Text Analytics
- (b) Azure Personalizer ✓
- (c) Azure Speech Service
- (d) Azure Translator

**Q154. In logistic regression, what type of output does the model produce?**

- (a) Probability scores ✓
- (b) Continuous numerical values
- (c) Categorical labels
- (d) Clusters of data points

**Q155. Which cloud service provides powerful tools for data analysis and visualization, including interactive dashboards?**

- (a) Tableau
- (b) Google Data Studio
- (c) Amazon QuickSight
- (d) Microsoft Power BI ✓

**Q156. What is the purpose of software installation tools like Anaconda and VSCode in AI development? (II)**

- (a) To demonstrate origami techniques
- (b) To install and manage necessary programming environments ✓
- (c) To showcase gardening skills
- (d) To create virtual reality experiences

**Q157. In natural language processing, what is the process of assigning grammatical tags to words in a text? (II)**

- (a) POS tagging ✓
- (b) Text normalization
- (c) Stemming
- (d) Tokenization

**Q158. What is type casting in Numpy used for?**

- |     |                                      |
|-----|--------------------------------------|
| (a) | Changing the shape of an array       |
| (b) | Changing the data type of an array ✓ |
| (c) | Changing the dimensions of an array  |
| (d) | Changing the elements of an array    |

**Q159. Which Azure service provides a platform for building and managing language understanding models, enabling intent recognition and entity extraction?**

- |     |                                |
|-----|--------------------------------|
| (a) | Azure Language Understanding ✓ |
| (b) | Azure Content Moderator        |
| (c) | Azure Text Analytics           |
| (d) | Azure Translator               |

**Q160. In machine learning, which algorithm creates a tree-like model of decisions and their possible consequences?**

- |     |                              |
|-----|------------------------------|
| (a) | Linear Regression            |
| (b) | Decision Tree ✓              |
| (c) | K-Means Clustering           |
| (d) | Support Vector Machine (SVM) |

**Q161. Which term in NLP refers to a curve that plots the performance of a model as a function of a certain parameter?**

- |     |                  |
|-----|------------------|
| (a) | Discourse curve  |
| (b) | Pragmatics curve |
| (c) | Learning curve ✓ |
| (d) | Syntax curve     |

**Q162. In neural networks, what are the two main steps involved in the training process? (II)**

- |     |                                            |
|-----|--------------------------------------------|
| (a) | Data Preprocessing and Feature Selection   |
| (b) | Forward and Backward Passes ✓              |
| (c) | Hyperparameter Tuning and Cross-Validation |
| (d) | Model Compilation and Evaluation           |

**Q163. Which Responsible AI practice ensures transparency and explainability of AI models and decisions? (II)**

- |     |                                   |
|-----|-----------------------------------|
| (a) | Model Accuracy                    |
| (b) | Explainability and Transparency ✓ |
| (c) | Performance Metrics               |
| (d) | Data Collection                   |

**Q164. In the classical machine learning pipeline, what step involves transforming raw data into a suitable format for analysis?**

- |     |                        |
|-----|------------------------|
| (a) | Feature crafting       |
| (b) | Data preprocessing ✓   |
| (c) | Modeling               |
| (d) | Testing and validation |

**Q165. In machine learning, which task involves predicting discrete class labels as output?**

- |     |                          |
|-----|--------------------------|
| (a) | Dimensionality reduction |
| (b) | Classification ✓         |
| (c) | Regression               |
| (d) | Clustering               |

**Q166. In object-oriented programming, what are instance variables?**

- |     |                                                                |
|-----|----------------------------------------------------------------|
| (a) | Variables that are shared among all instances of a class       |
| (b) | Variables that store data unique to each instance of a class ✓ |
| (c) | Variables that belong to the class itself                      |
| (d) | Variables used for method definition                           |

**Q167. What does multiple inheritance in OOP involve?**

- |     |                                           |
|-----|-------------------------------------------|
| (a) | Inheriting from a grandparent class       |
| (b) | Inheriting from a child class             |
| (c) | Inheriting from a single parent class     |
| (d) | Inheriting from multiple parent classes ✓ |

**Q168. Which service enables you to manage identity and access for Azure AI applications by providing secure authentication and authorization? (II)**

- |     |                          |
|-----|--------------------------|
| (a) | Azure Key Vault          |
| (b) | Azure Security Center    |
| (c) | Azure Firewall           |
| (d) | Azure Active Directory ✓ |

**Q169. Which evaluation metric calculates the ratio of correctly predicted positive observations to the total predicted positives?**

- (a) Accuracy
- (b) Precision ✓
- (c) F1-score
- (d) Recall

**Q170. How can you convert a user's input into a numerical data type in programming?**

- (a) By performing a bitwise operation
- (b) Through the input() function and type casting ✓
- (c) Using the stringify() function
- (d) By using the floatify() method

**Q171. In cloud services, what are used as credentials to securely authenticate and access resources, and are often required to be protected and rotated regularly?**

- (a) API keys
- (b) Account keys ✓
- (c) Access tokens
- (d) Encryption keys

**Q172. In logistic regression for multiclass classification, what type of decision boundary is typically learned by the model? (II)**

- (a) Exponential decision boundary
- (b) Linear decision boundary ✓
- (c) Non-linear decision boundary
- (d) Constant decision boundary

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Correct answers are highlighted in green ✓